

01 Serverless Computing

What is Serverless Computing?

Traditional vs Serverless Architecture

Traditional Server-Based Computing:

- Purchase/rent physical servers or virtual machines
- Manage operating systems, patches, updates
- Handle capacity planning and scaling
- Pay for idle time when servers aren't fully utilized
- Responsible for infrastructure maintenance and monitoring

Serverless Computing:

- No server management required
- Code runs in ephemeral compute containers
- Automatic scaling based on demand
- Pay only for actual compute time consumed
- Infrastructure abstraction - focus purely on code

Key Serverless Principles

1. No Server Management

- Cloud provider handles all infrastructure
- Automatic provisioning and de-provisioning
- Zero administrative overhead

2. Event-Driven Execution

- Functions execute in response to events
- Stateless execution model
- Automatic resource allocation per request

3. Fine-Grained Billing

- Pay per execution and duration
- No charges for idle time
- Cost scales directly with usage

4. Automatic Scaling

- Scales from zero to thousands of concurrent executions
- No capacity planning required
- Built-in high availability